1. Let A be a set of vectors in $\mathbf{R}^n$ and $\mathbf{B}$ be a matrix (not necessarily square).
Determine following statement whether it is true or false and do some brief reason.

1.1. $\text{perp}(A) = \text{perp}(\text{span}(A))$

1.2. $\text{perp}(\text{perp}(A)) = A$

1.3. If $|A| = n - 1$, then $\text{span}(A)$ can **never** be $\mathbf{R}^n$.

1.4. If $|A| = 2n + 1$, then $\text{span}(A)$ will **always** be $\mathbf{R}^n$.

1.5. If $\text{row}(\mathbf{B}) = \text{col}(\mathbf{B})$, then $\mathbf{B}$ is symmetric.

1.6. If $\mathbf{B}^T = -\mathbf{B}$, then $\text{row}(\mathbf{B}) = \text{col}(\mathbf{B})$.

1.7. If $\mathbf{B}^T = -\mathbf{B}$, then $\mathbf{B}x \perp x$ for all possible vectors x.

1.8. All matrices that have their nullspace equal to $\{\mathbf{0}\}$ have no free columns.

2. Find a matrix (if possible) which satisfies each condition below.

   2.1. a 3×3 matrix whose nullspace is a point.

   2.2. a 3×3 matrix whose nullspace is a line.

   2.3. a 3×3 matrix whose nullspace is a plane.

   2.4. a 2×2 matrix whose nullspace and column space both lie on the span of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

3. Let $\mathbf{A} = \begin{bmatrix} 0 & 10 & -1 & -1 \\ -1 & 2 & 99 & 0 \end{bmatrix}$ and $\mathbf{B}$ be the reduced row echelon form of $\mathbf{A}$, i.e., $\mathbf{B} = \mathrm{RRE}(\mathbf{A})$

   3.1. Let $S = \mathrm{row}(\mathbf{B})$ (the row space of $\mathbf{B}$). Write S as a **span of a finite set of vectors**.

   3.2. Let $S = \mathrm{col}(\mathbf{B})$ (the column space of $\mathbf{B}$). Write S as a **perp of a finite set of vectors**.

3.3. Let S be the left null-space of $\mathbf{B}^T$.

Write S as a **span of vectors** whose **entries are all integers.**

3.4. Give an example of a vector in $\mathbf{R}^4$ that belongs to both row($\mathbf{A}$) and nul($\mathbf{A}$)

4. Let $\mathbf{A} = \begin{bmatrix} 2 & 4 & -2 & 4 \\ 3 & 6 & 3 & -6 \end{bmatrix}$

4.1. Find an explicit span description for nul($\mathbf{A}$).

4.2. Find an implicit perp description for row($\mathbf{A}$).

5. Consider a set of vectors in each sub-problem. Determine whether it is **linearly independent** or not. If not, write one vector in the set as a linear combination of the others.

5.1. $\left\{ \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} \right\}$

5.2. $\left\{ \begin{bmatrix} -1 \\ 2 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -5 \\ 5 \\ -3 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 4 \\ -1 \\ 9 \\ -4 \end{bmatrix} \right\}$

6. Consider a matrix $\mathbf{A}$ where some entries are missing $\mathbf{A} = \begin{bmatrix} -4 & 2 & -2 & 6 \\ 3 & -1 & 1 & 0 \\ 8 & x & 3 & y \end{bmatrix}$.

If we know that the basis of $\text{row}(\mathbf{A})$ contains exactly 2 vectors. Answer these following questions.

6.1. Nullity of $\mathbf{A}^T$

6.2. Basis of $\text{row}(\mathbf{A})$

6.3. The smallest set S such that $\text{perp}(S) = \text{nul}(\mathbf{A})$

6.4. The smallest set U such that $\text{span}(U) = \text{nul}(\mathbf{A}^T)$

7. One of the two sets in Question 5. seems to be linearly dependent. Find a basis for the span of that set.

8. Consider each statement below about the rank r of an $n \times m$ matrix $\mathbf{A}$.
Determine whether it is true or false.

8.1. If $r < n$, the columns of $\mathbf{A}$ are dependent.

8.2. If $r < n$, the vectors from rows of $\mathbf{A}$ can never be a basis for any subspace.

8.3. If $m = n = r$, then A is invertible.